Vaibhav Krishna

Embedded Systems | IoT | Electronics | Mechatronics Engineer

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Professional Profile

Mechatronics and electronics-focused engineer with practical experience in ESP32-based IoT devices, embedded firmware, industrial sensing, MQTT telemetry, ThingsBoard dashboards, PCB design, wiring architecture, and SolidWorks enclosure development. Experienced in converting real-world sensor measurements into structured data for monitoring, analytics, field trials, and product-ready deployments.

Core Technical Skills

Embedded / Firmware	ESP32, ESP32-S3, STM32, Arduino IDE, C/C++, I2C, SPI, UART, ADC, EEPROM, OTA, deep sleep, calibration
IoT / Cloud	MQTT, ThingsBoard, dashboards, RPC control, shared attributes, WiFi portals, LoRa, telemetry pipelines
Electronics / Sensors	Current, vibration, NTC, SHT31, MPU6050, ADXL345, SEN66, SPS30, ENS160, ADS1115, 4-20 mA sensors
Design / Software	Multi-layer PCB design, SolidWorks enclosures, DFM, Java, C#, Python, JavaScript, Angular, React, Spring Boot, SQL, Docker

Selected Engineering Projects

CLEAR Environmental Monitoring Platform

- Developed cloud-connected environmental monitoring functionality for CO2, PM, VOC index, NOx index, temperature, humidity, and noise.
- Integrated SEN66, SPS30, and INMP441 sensors with ESP32-S3 hardware, MQTT telemetry, dashboard visualisation, alerts, and deployment-ready optimisation.

Current Sensor Utilisation System

- Built an ESP32-based machine-utilisation monitoring system using RMS current measurement, ThingsBoard telemetry, and two-way dashboard control.
- Implemented remote threshold and sampling-rate updates through RPC and shared attributes for industrial asset monitoring.

StrucSense Structural Monitoring Device

- Developed a wireless concrete monitoring device measuring temperature, humidity, tilt, acceleration, and battery voltage.
- Implemented MPU6050 pitch/roll calculation, baseline correction, SHT31 sensing, WiFi/LoRa communication, and low-power logic.

Motor, Vibration and Flow Monitoring Rig

- Worked on a VFD-driven motor and pulley rig for vibration, current, flow, and temperature data collection to support anomaly detection and AI training workflows.

SolidWorks Enclosure Design

- Designed multiple enclosures for CLEAR and smaller connected-device projects, focusing on mounting, cable routing, manufacturability, sensor exposure, accessibility, and product identity.

Professional Experience Direction

Embedded / IoT / Electronics Engineering: Developing IoT hardware and firmware for industrial sensing, environmental monitoring, current monitoring, connected analytics platforms, and product-ready field deployments.

Product Development and Documentation: Experience across sensor selection, embedded coding, PCB and wiring design, enclosure development, cloud dashboards, test reports, calibration reports, and technical documentation.

Education and Certifications

MSc Cyber Security, Nottingham Trent University - Distinction | B.Tech Information Technology

ISO 27001:2013 | AWS Technical Essentials | Azure administration preparation | PLC and industrial automation coursework | Full-stack development experience